

ECTE 250
DETAILED DESIGN REPORT
DELIVERABLE 2

By: Arc Tech

(Team Members:

Muhammad Haaziq (8927133)

Faaz Ali Sayyed (8943564)

Daniel Koshy (8938799)

Mohammed Abdul Rahman (9070734)



UNIVERSITY
OF WOLLONGONG
IN DUBAI

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1. Executive Summary

This report presents the detailed design of NurdleDNA: an IoT-enabled, in-line water monitoring and capture unit designed to detect and intercept plastic pellets (“nurdles”) in industrial water flows. The product combines a vision chamber, UV illumination, an AI detection module, and a servo-controlled diverter valve that routes suspected pellet events to a capture cartridge mounted on a load cell for mass measurement. A gas sensor provides a parallel safety layer for volatile organic compound (VOC) detection above the water chamber. The proposed design solution uses a multi-processor approach: an NVIDIA Jetson Nano (or equivalent) runs AI vision and the dashboard, while an Arduino reads analog sensors and controls real-time actuators (servo, indicators), communicating via USB serial. This design aligns strongly with the ECTE250 IoT, “Reduce, Reuse, Recycle” theme by preventing microplastic release and enabling measurable capture for reporting and compliance. Key outcomes planned for the remainder of the session include: (1) complete state-machine simulation verified against the state chart (S1-W10 lab requirement), (2) TinkerCAD prototype demonstration, then (3) breadboard and perfboard prototype build, test, and innovation-fair-ready demo.

2. Introduction

2.1 Background and Problem

Microplastics are plastic particles smaller than 5 mm. A major and preventable contributor is pre-production plastic pellets (“nurdles”), which are the raw feedstock for many plastic products. Nurdles are handled in very large volumes across polymer plants, logistics yards, bagging lines, ports, and industrial zones. During routine operations—bag rupture, silo transfer, loading/unloading, valve leakage, filter bypass, sweeping, and washdown—pellets can escape and enter stormwater drains, sumps, and outfalls. Once released, plastics persist for decades: they do not biodegrade under normal environmental conditions, but fragment into smaller micro- and nano-plastics, increasing mobility and bioavailability.

This project expands the “nurdle spill” challenge into a broader industrial and household safety context. Liquid and drainage networks often carry multiple hazards simultaneously, including microplastics, hydrocarbons, volatile chemicals,

and flammable or toxic gases in confined spaces. A practical solution must therefore be able to detect and manage solids (microplastics), liquids (chemical and hydrocarbon contamination), and gases (VOCs and leak indicators) in a single, deployable unit. Nurdle DNA is proposed as an inline “industrial firewall” for effluents: it detects abnormal contamination signatures in real time, physically isolates or diverts flow, captures evidence, and generates data.

2.2 Environmental and Health Impacts

Nurdles resemble food and are ingested by fish, seabirds, and turtles. Ingestion can cause internal injury, gut blockage, false satiety, reduced growth, and impaired reproduction. Plastics also adsorb pollutants (e.g., persistent organic contaminants and metals), meaning ingested pellets may transport toxins into organisms. Over time, microplastics move through food webs and raise concerns for human exposure through seafood and water.

In addition to marine impacts, microplastics are increasingly reported in drinking and bottled water systems. Mechanical wear, heat, ultraviolet exposure, and production processes can introduce plastic particles into consumable liquids. This creates a second motivation for detection technology: beyond environmental compliance, there is rising demand for in-line quality assurance in food and beverage handling, and potentially in smart homes (e.g., household water filtration and plumbing leak monitoring).

2.3 Industrial Safety Risks and Multi-Hazard Reality

Industrial drainage pathways can become hazardous due to flammable vapor accumulation, solvent leaks, hydrocarbon seepage, or chemical washdown. Plastic pellets can carry static charge and may coexist with VOCs, increasing fire or explosion risk in confined drainage systems. Hydrocarbon contamination can indicate leaks from fuel, lubricants, or process lines. Certain chemical leaks (acidic or alkaline cleaning fluids) can harm workers, damage equipment, and create regulatory violations. Conventional “dumb filters” cannot warn operators when these risks emerge, and manual inspection is too slow.

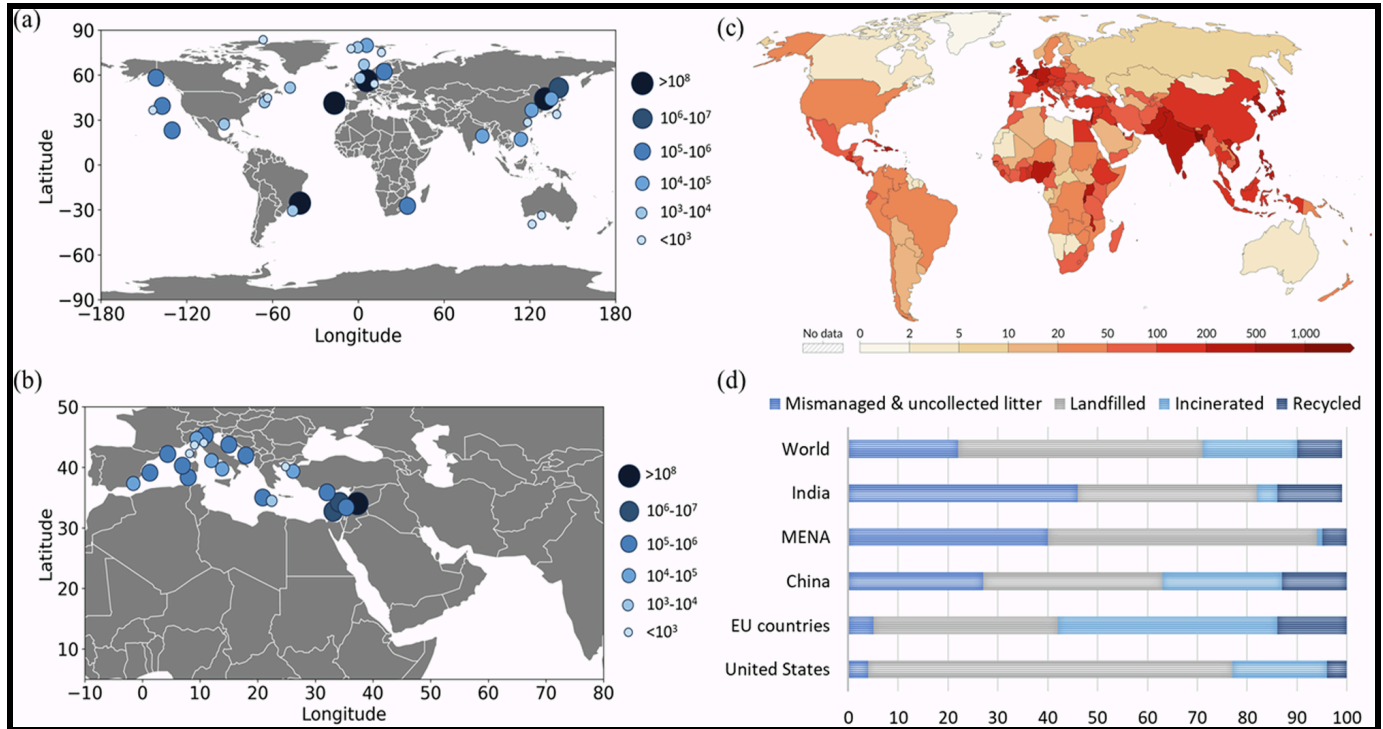


Figure 1: The density of microplastics Pollution Globally and in the MENA Region related to urban and industrial sites locations and to the intensity of their activities.

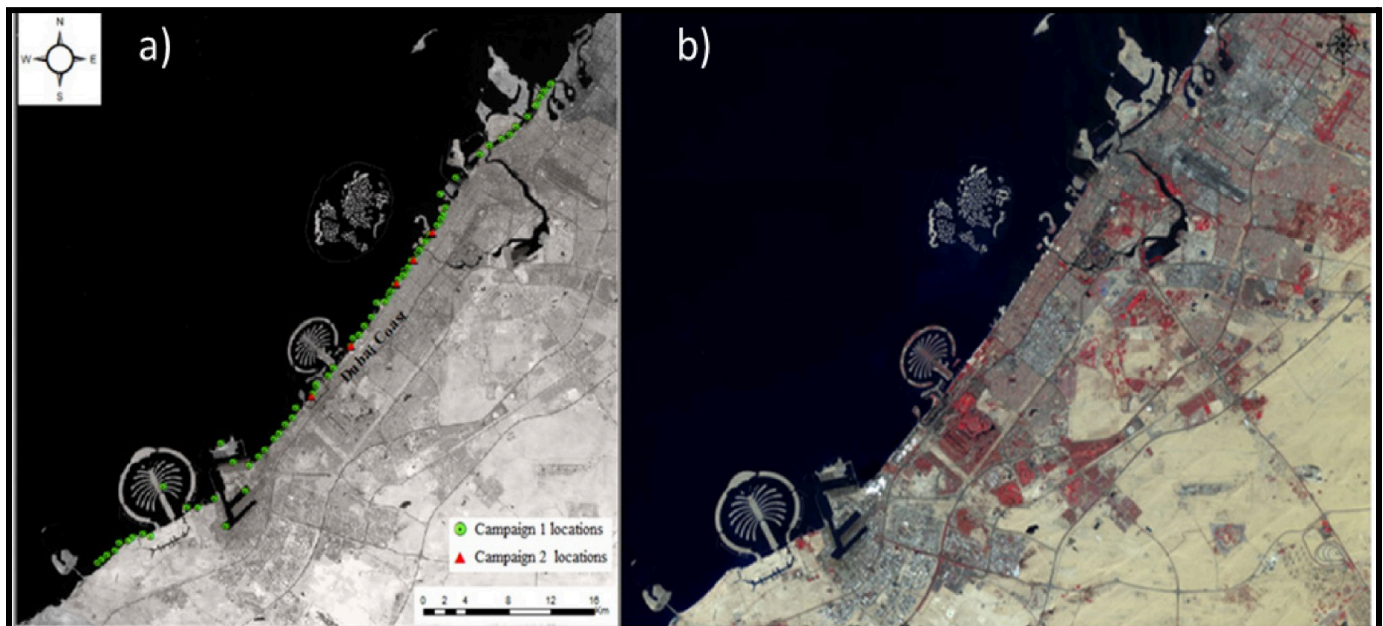


Figure 2: Evaluating Microplastic Pollution Along the Dubai and Dubai Coast and Coastal waters MPs sampling locations along the coast of Dubai (a) and Sentinel-2 image (b) of the study area acquired

2.4 Limitations of Current Solutions/Competitive Edge

Existing approaches typically fall into one of four categories:

- (a) Manual housekeeping and spill response: Highly dependent on human behavior, with delays and inconsistency.
- (b) Passive traps/filters: low cost but prone to clogging and overflow; provide no proof of what escaped.
- (c) Laboratory identification (FTIR/Raman/microscopy): accurate but slow, costly, and not continuous.
- (d) Alarm-only monitoring: can detect changes but does not automatically stop discharge or preserve evidence.

In shared industrial facilities, accountability is a recurring problem. When an incident occurs at a shared bay, contractors may deny responsibility, and site owners often bear cleanup and reputational costs. Without time-stamped evidence and material signatures, spills become “orphan pollution.”

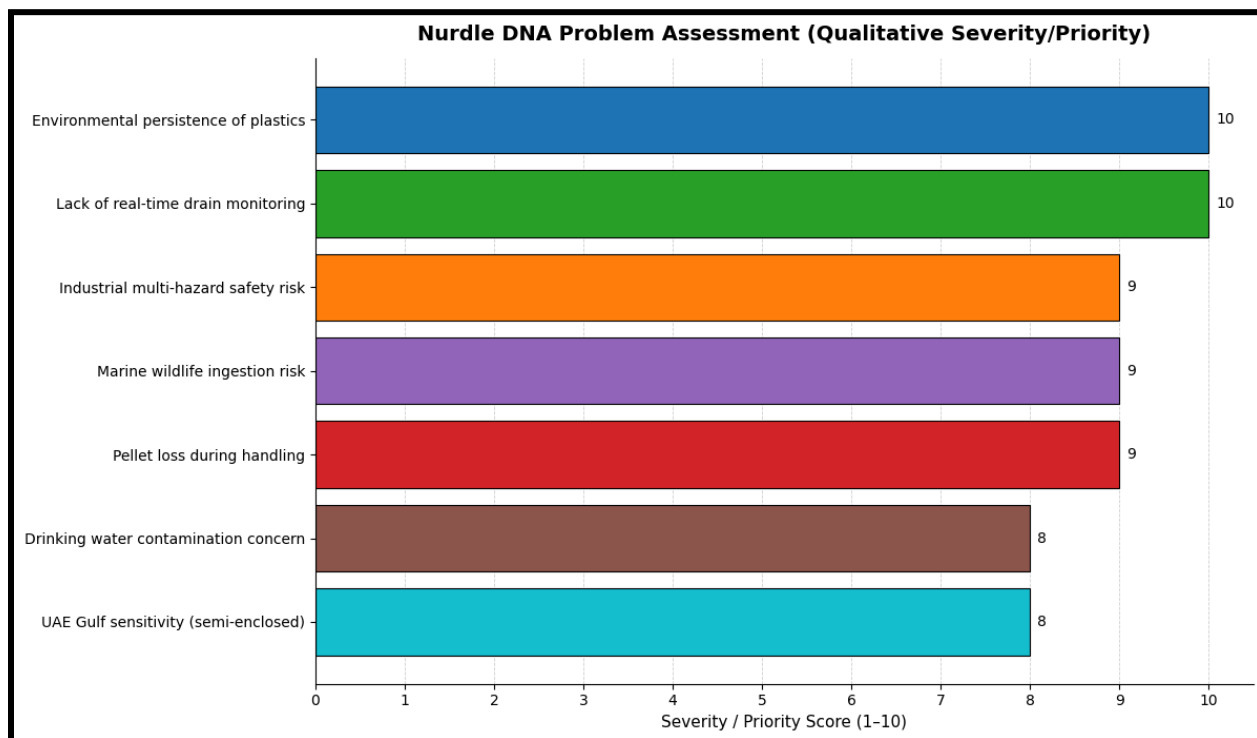


Figure 3 : Graph depicting the Nurdle DNA Problem assessment in a score of 10

2.5 Proposed Solution

A retrofit-friendly device is required that can be installed inline within a pipe section or as a bypass loop to: (i) detect microplastics and abnormal particulate loads in liquid streams; (ii) detect hydrocarbon and harmful chemical leakage indicators; (iii) detect gas leakage or VOC accumulation in drainage headspace; (iv) automatically isolate or divert flow to prevent downstream release; (v) capture physical evidence for chain-of-custody; and (vi) create automated compliance reports and alerts. Nurdle DNA is designed to address this integrated requirement.

3. Design

3.1 System Concept

Nurdle DNA is an inline, retrofittable module that attaches to a pipe or forms part of an industrial machine's discharge pathway. The device continuously monitors liquid flow for microplastics and other contamination signatures, monitors headspace for gas leakage, and applies automated containment logic when risk thresholds are exceeded. It logs all contamination events in a database for later use.

3.2 Detect–Classify–Actuate–Capture–Report Loop

The core architecture follows five steps:

1. Detect - Sensors continuously monitor the water flow and surrounding air for microplastics, chemical anomalies, and hazardous gases.
2. Classify - The system analyzes sensor data against site-specific baselines to reliably distinguish real hazards from normal water fluctuations.
3. Actuate - The device triggers a valve to shut off or divert flow into a holding line/tank.
4. Capture - A removable evidence cartridge traps solids and allows sampling for confirmation.
5. Report - The event is immediately time-stamped and securely uploaded to a centralized cloud dashboard to alert operators and maintain an audit trail for compliance

3.3 Design Requirements

Key requirements guiding the concept include: continuous monitoring, fast response time, fail-safe behavior, tamper-resistant evidence capture, retrofit installation, minimal pressure drop, low maintenance, secure connectivity, and scalable cost per unit.

3.4. Component Design Description

3.4.1 Mechanical and Flow Architecture

The device is intended to be installed as a spool piece or bypass loop. The flow path contains (i) a sensing zone, (ii) an actuation zone (valve/diverter), and (iii) an evidence capture zone (cartridge). For prototype demonstration, a closed sampling loop with a peristaltic pump is used to represent continuous intake from a drain or pipe. In industrial deployment, a bypass loop is preferred because it allows monitoring without fully restricting the main pipe diameter and enables maintenance without halting operations.

3.4.2 AI Vision Layer for Forensic Attribution

A small camera (or optical sensor array) can image particles passing through a transparent section. Edge inference estimates particle count, pellet-like morphology, and basic color/fluorescence cues. This does not replace lab spectroscopy, but provides practical “signature” evidence for attribution. In the prototype, this function can be simulated or implemented using a compact edge compute board; in production, a lower-power edge module may be used for cost and reliability.

3.4.3 UV/Fluorescence Option

Many pellets include additives that can fluoresce under UV. A UV LED and photodiode can capture relative fluorescence intensity, supporting simple classification. This feature strengthens the “DNA” concept by adding a measurable fingerprint dimension.

3.4.4 Gas Leakage Detection Subsystem

A gas sensor module monitors VOCs or other relevant gases in the headspace above the liquid. In drainage systems, gas accumulation can occur even when liquid flow appears normal. The device treats gas exceedance as an HSE hazard and trigger an alarm.

3.4.5 Embedded Control and Decision Logic

An embedded microcontroller reads sensors at defined sampling intervals, applies smoothing filters, and calculates a confidence score for contamination states. States are defined as:

- NORMAL: within baseline envelope.
- WARNING: abnormal trend; logging and alerting.
- ALARM: exceedance; latching action required.

The latching behavior ensures the system does not “oscillate” open/closed during unstable conditions; reset requires authorized acknowledgement.

3.4.6 Interlock: Automatic Shutoff or Diversion

The core differentiator is actuation. In the prototype, a servo pinch valve clamps tubing to stop flow. In industrial deployment, an actuated valve (pinch, ball, gate, or diverter valve) isolates discharge or diverts it to a holding tank. The actuation requirement is: fast closure, reliable sealing, safe fail mode, and compatibility with the site’s pipe material and pressure.

3.4.7 Evidence Collection Cartridge and Load Measurement

A removable cartridge contains a mesh filter sized to capture pellets and targeted particle ranges. The cartridge can be sealed after removal to preserve evidence. A load cell beneath the cartridge can quantify captured mass, providing objective incident magnitude. This enables reporting beyond “alarm triggered” and supports trend analysis (e.g., repeated small losses).

3.4.8 Local Interface and Alarms

The device includes an OLED display and status LEDs/buzzer. At minimum it shows: device ID, state (NORMAL/WARNING/ALARM), captured mass estimate,

and last event time. Local alarms are necessary because connectivity may fail; the device must still warn operators on site.

3.4.9 IOT and Data Collection

A Wi-Fi or cellular module transmits event logs to a cloud dashboard. Data includes sensor values, valve actions, timestamps, and evidence mass. The dashboard supports compliance reports, maintenance planning, and multi-site monitoring. Secure protocols and authentication are assumed in the design.

3.5 TinkerCad Design

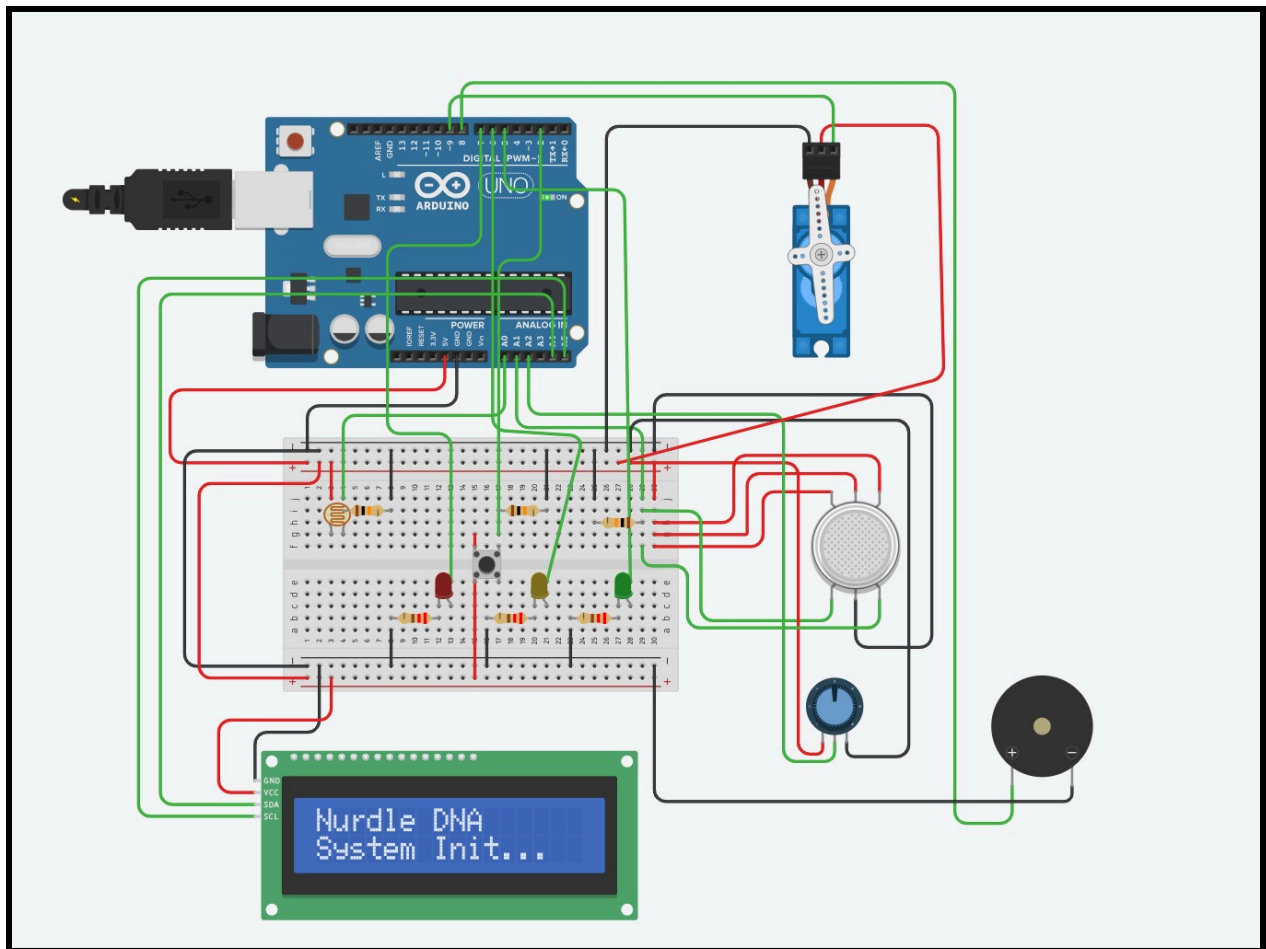


Figure 4: TinkerCAD Circuit Model

Evidence Of Testing

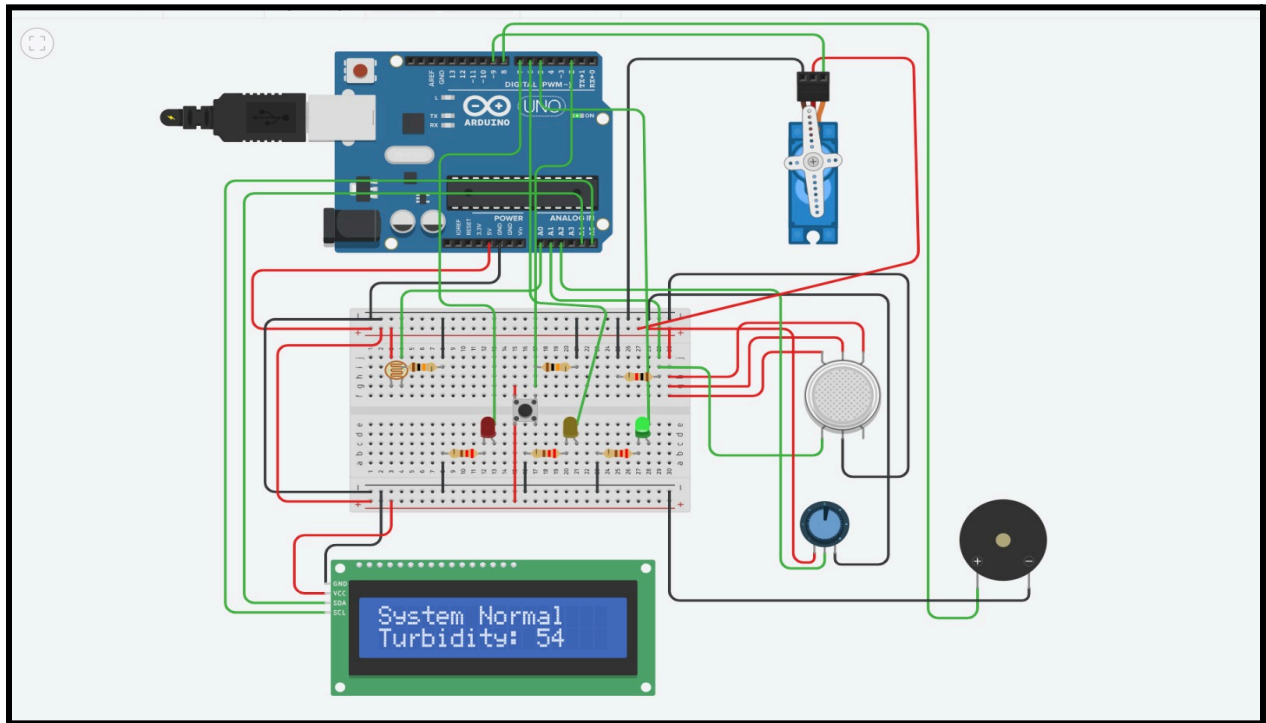


Figure 5: Regular Water Flow (When no nurdles are detected)

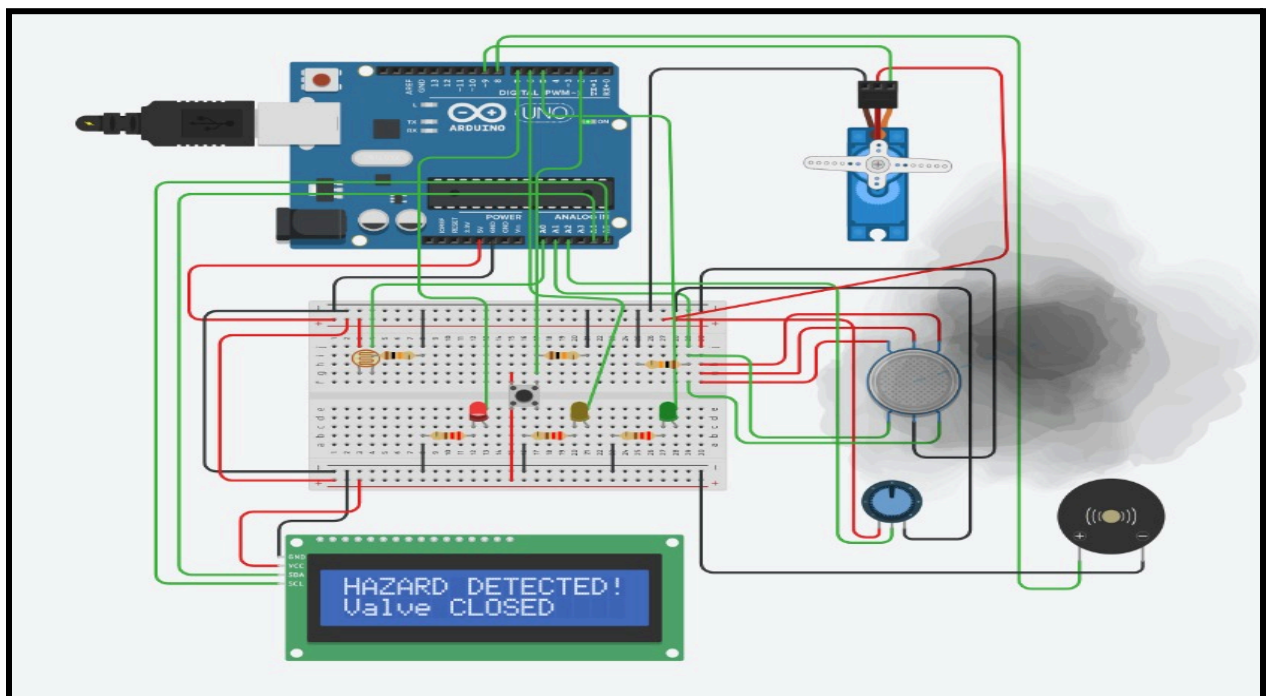


Figure 6: Gas Leak Detected

Arduino Code:

(Included in the Appendix)

Design Model of the Prototype

Design Sketch

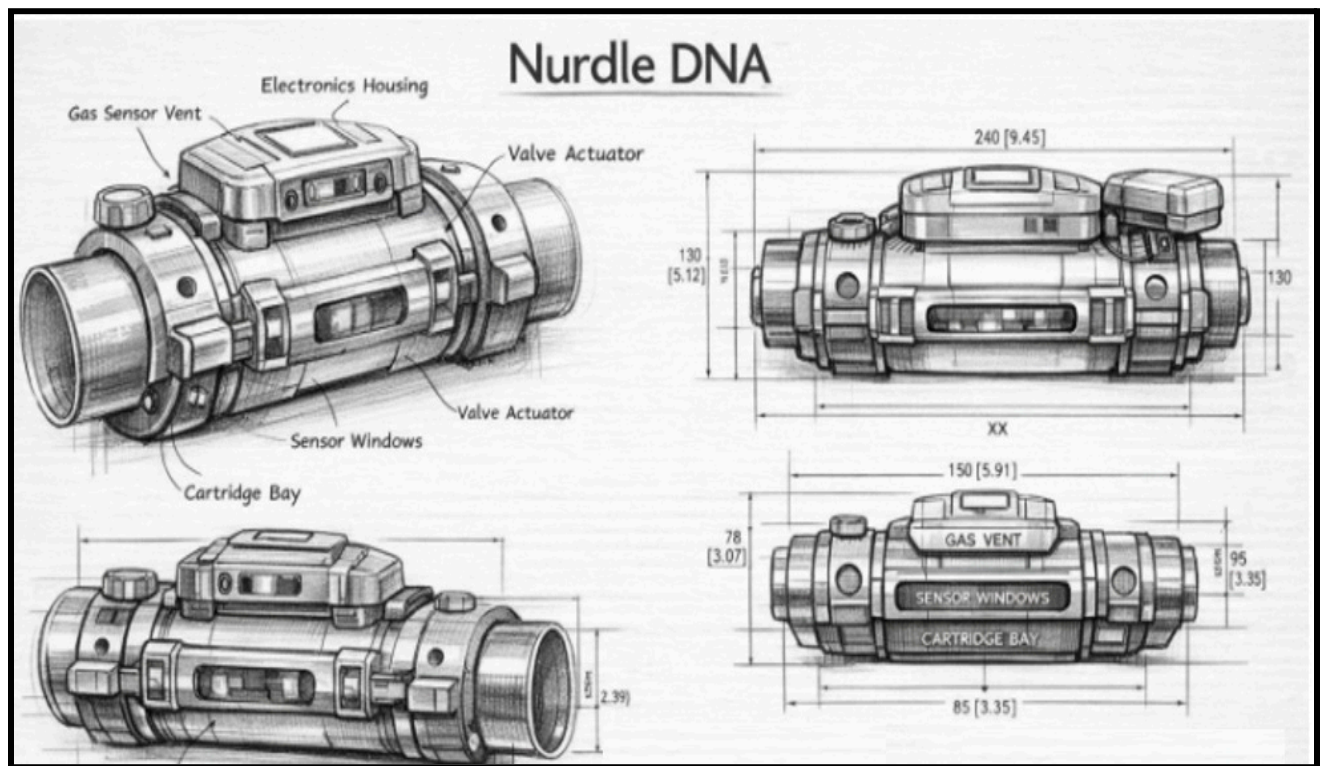


Figure 7: Design Sketch

3D Design Model



Figure 8: 3D Model Design Sketch

The Design Model illustrates how the prototype looks in Real Application Scenarios

3.6 State Machine

To make the system behaviour deterministic and safe, NurdleDNA uses a Moore finite state machine (FSM). In a Moore machine, the outputs depend only on the current state. The FSM receives three decision inputs (WARN, CRIT, RST) and produces five outputs in the order (VALV, Green, Yellow, Red, Buzzer) where VALV = 1 means the valve is open / normal flow allowed.

Inputs

- **WARN:** warning-level condition. This represents “something suspicious” but not confirmed (for example, low-confidence nurdle detections or early signs in the sensing logic).
- **CRIT:** critical condition. This represents a confirmed event requiring immediate protective action (for example, confirmed nurdle detection, or gas

leak detection).

- RST: operator reset push-button used to acknowledge and clear the alarm latch.

Outputs

- VALV: valve command. 1 = open (normal), 0 = closed (protective).
- G: green status LED
- Y: yellow status LED
- R: red status LED
- B: buzzer

States

Table 1: NurdleDNA States

State ID	Name	What it represents in the system
S0 (11110)	INIT	Startup self-test / lamp test before entering monitoring
S1 (11000)	SysOk	Normal monitoring and operation (system healthy)
S2 (10100)	Causn	Caution mode (warning-level condition detected)

S3 (00011)	ALRM	Critical alarm mode (protective action + latched)
S4 (10000)	RSTIN	Reset/clearing sequence after operator acknowledgement

S0 – INIT (Startup test)

This is the first state after power is applied. The aim is to show a quick “system alive” indication by turning on the LEDs together (buzzer off) while keeping the valve open. The FSM does not stay here long and it moves automatically into normal monitoring.

S1 – SysOk (Normal monitoring)

This is the steady running state. The valve stays open and the green LED indicates normal operation. The system remains here as long as there is no warning or critical condition. If a warning appears, it goes to caution and if a critical event occurs, it escalates immediately to alarm.

S2 – Causn (Caution / warning)

This state represents an abnormal condition that is not yet severe enough to trigger protective shutdown. The valve stays open but the yellow LED warns the operator. If the warning clears, the FSM returns to SysOk. If the condition becomes critical, it escalates immediately to ALRM.

S3 – ALRM (Critical alarm, latched)

This is the protective state. The valve is closed and the red LED and the buzzer are active. The key feature is that ALRM is latched i.e. even if the critical input later disappears, the FSM stays in ALRM until the operator presses reset. This prevents the system from silently returning to normal and ensures the event is acknowledged.

S4 – RSTIN (Reset/clearing)

This is a short clearing state entered when the operator presses reset. It indicates the alarm has been acknowledged and the system is preparing to resume monitoring. RSTIN returns to SysOk when RST is released.

State Diagram

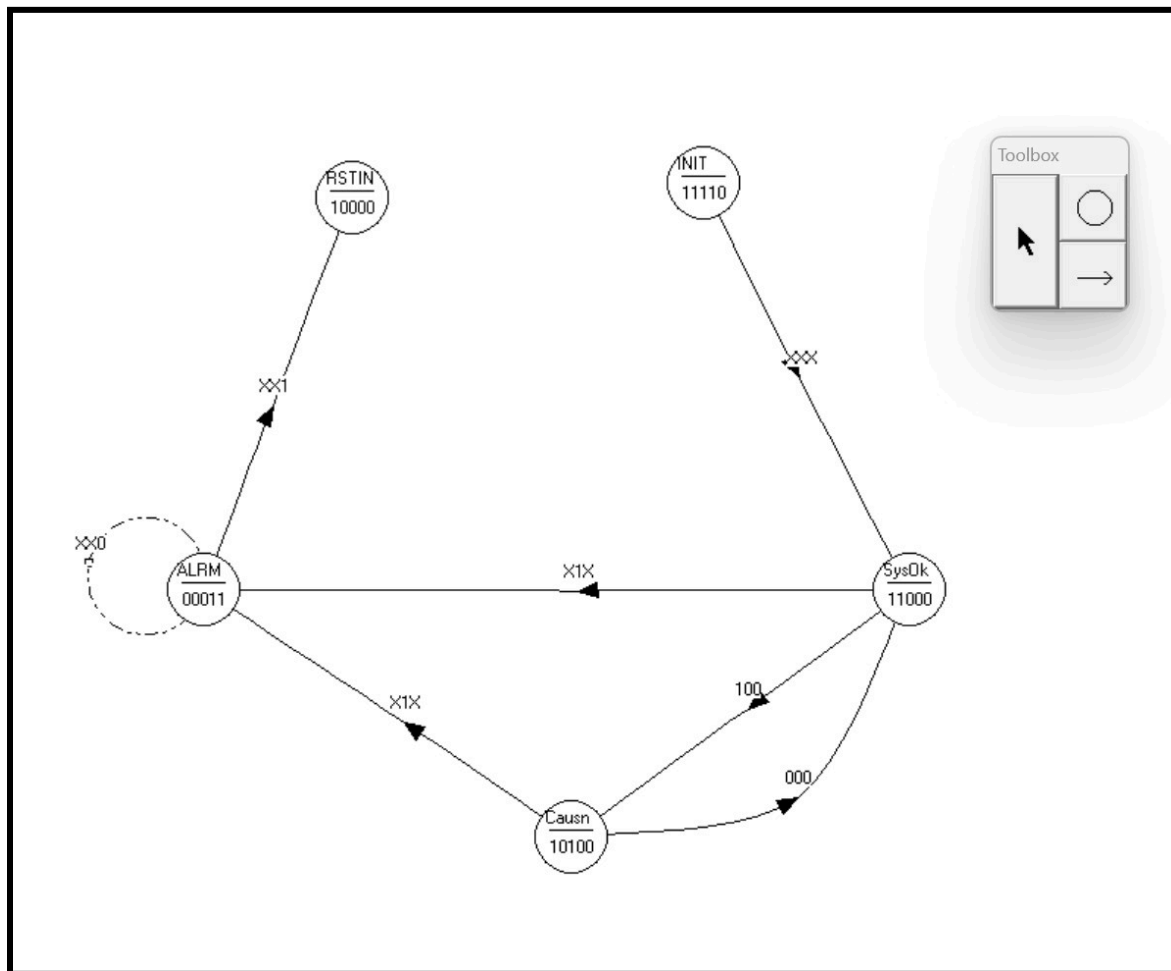


Figure 9: State Diagram

State Table

Transition Table								
	000	001	010	011	100	101	110	111
SysOk	X	X	ALRM	ALRM	Causn	X	ALRM	ALRM
Causn	SysOk	X	ALRM	ALRM	X	X	ALRM	ALRM
ALRM	ALRM	RSTIN	ALRM	RSTIN	ALRM	RSTIN	ALRM	RSTIN
RSTIN	X	X	X	X	X	X	X	X
INIT	SysOk	SysOk	SysOk	SysOk	SysOk	SysOk	SysOk	SysOk

State and Transition Table											
	Qi	Output		000	001	010	011	100	101	110	111
SysOk	000	11000	000	xxx	xxx	010	010	001	xxx	010	010
Causn	001	10100	001	000	xxx	010	010	xxx	xxx	010	010
ALRM	010	00011	010	010	011	010	011	010	011	010	011
RSTIN	011	10000	011	xxx	xxx	xxx	xxx	xxx	xxx	xxx	xxx
INIT	100	11110	100	000	000	000	000	000	000	000	000
X	101	XXXXX	101	xxx	xxx	xxx	xxx	xxx	xxx	xxx	xxx
X	110	XXXXX	110	xxx	xxx	xxx	xxx	xxx	xxx	xxx	xxx
X	111	XXXXX	111	xxx	xxx	xxx	xxx	xxx	xxx	xxx	xxx

Figure 10: State Table And Transition Table

State Expressions

```

J0=(~CRIT*~q2*~q1)+(WARN*q1)
K0=(1)
J1=(CRIT*~q2)
K1=(0)
J2=(0)
K2=(1)
D0=(WARN*q1)+(~CRIT*~q2*~q1*~q0)
D1=(q1)+(CRIT*~q2)
D2=(0)
VALV=(q1*~q0)
LedG=(q1*~q0)+(q2)
LedY=(~q1*q0)+(q2)
LedR=(~q1*~q0)
BUZZ=(~q1)+(q0)

```

Figure 11: State Expression

Multisim Circuit

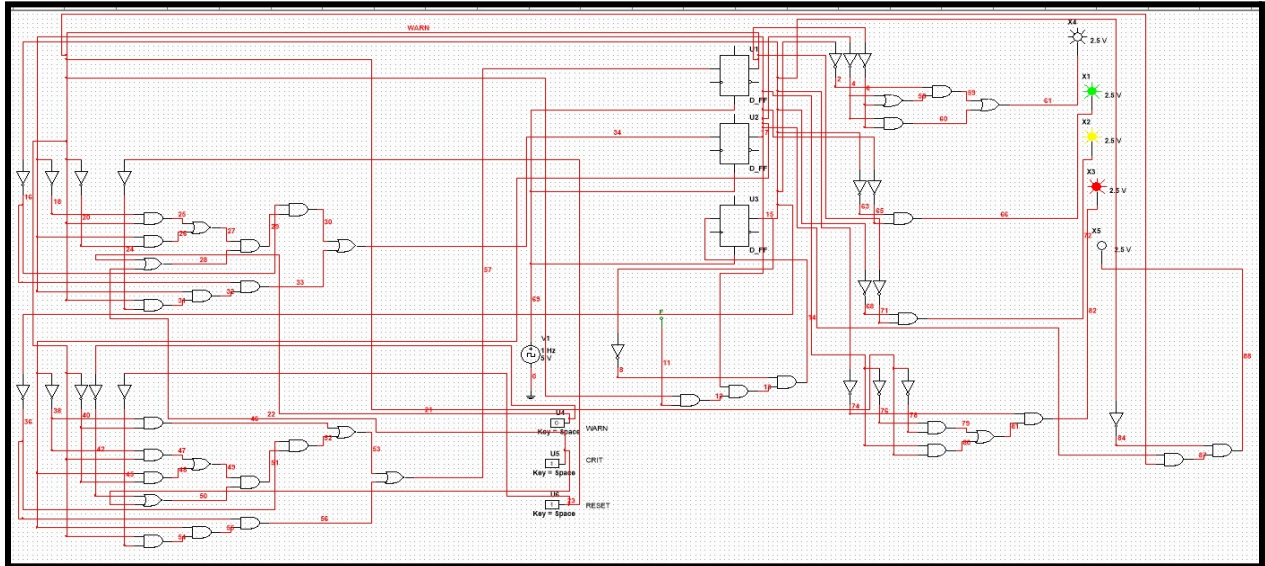


Figure 12: Multisim Circuit

Block Chart

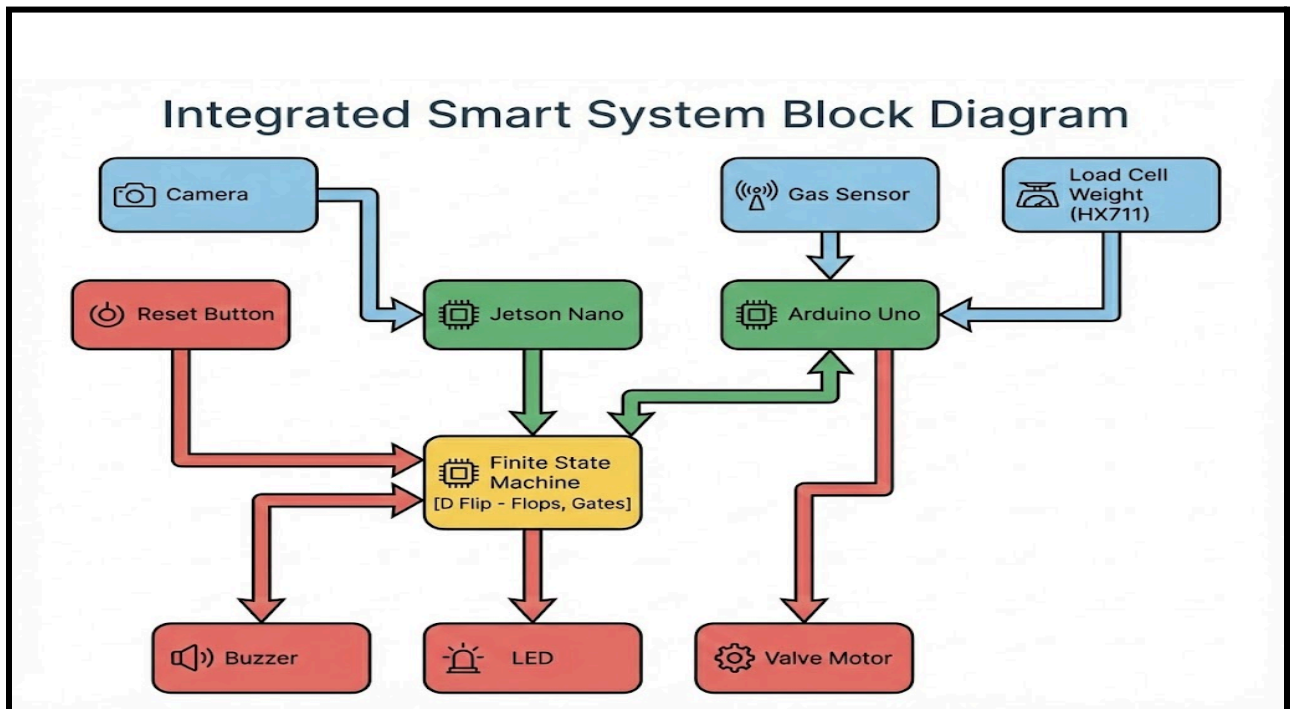


Figure 13: Component Block Chart

4. Alignment

Our Nurdle DNA design directly aligns with the core requirements: continuous inline monitoring, rapid detection, automatic containment/diversion, evidence capture, and logged reporting for compliance and accountability. Key constraints addressed include retrofit installation (spool/bypass section without major pipe redesign), low maintenance (removable cartridge and serviceable sensors), site variability (baseline calibration to handle different turbidity backgrounds), and fail-safe operation. Complexity is demonstrated through the combined use of AI vision, a hardware interlock state machine, multi-sensor measurement and safety triggers, multi-processor serial communication, and local LED/buzzer indicators.

Here are some of the key changes we made to the project in this updated version.

- We upgraded the detection method from a basic optical/turbidity approach to a camera-based AI system on the Jetson Nano that detects nurdles as individual objects, supported by UV illumination for clearer visibility.
- We redesigned the computing architecture into a two-processor setup, where the Jetson runs AI and logging while the Arduino handles analog sensor readings and real-time valve control.
- We strengthened the capture measurement feature by placing the removable mesh cartridge on a load cell (via the HX711) so the system can report the mass of pellets collected as a measurable KPI.
- We added a gas/VOC sensing layer using an MQ-series sensor and defined a separate hazard alarm response to improve safety monitoring.
- We standardized the control logic using the Boole–Deusto, five-state Moore FSM (INIT, SysOk, Causn, ALRM, RSTIN), including a latched alarm state that remains active until the operator resets the system.
- We improved traceability by adding proof-image and event logging, and we clarified the signal interfaces between the Jetson, FSM, Arduino, sensors, and the valve actuator to simplify testing and demonstration.

5. Testing

Testing will be conducted in two stages. In the simulation stage, sensing thresholds will be validated using synthetic datasets, false-trigger rates will be modelled, and control-logic timing will be verified. Gas and chemical sensing channels will be evaluated using simulated sensor response curves and fault-injection tests to confirm safe-state behaviour. Breadboard testing will then validate each subcomponent independently. Finally, end-to-end validation will be performed using a closed-loop test rig with timed spill events to confirm detection accuracy, diversion/closure response time, evidence capture, alarm functionality, and correct event logging to the dashboard.

6. Plan

6.1 Deliverables Schedule and Planned Work

Deliverable 3 – Design Simulation

The next step is to complete the full state transition table and output table for the Moore FSM, derive the next-state logic equations, and implement the circuit in Multisim using flip-flops and logic gates. Verification will be done using structured test vectors (normal → warning → alarm latch → reset recovery), with screenshots and simulator files compiled into the required short technical report.

Deliverable 4 – TinkerCAD Prototype

A simplified prototype will be built in TinkerCAD to demonstrate the same functional chain: WARN/CRIT inputs driving the FSM behaviour and producing valve/indicator outputs. The prototype will include representative sensor blocks and servo control behaviour, supported by clear screenshots and a short results summary.

Deliverable 5 – Breadboard Prototype

The system will be migrated from simulation to a physical breadboard prototype. Work will focus on stable wiring, correct power distribution, sensor bring-up (gas sensor and load cell via HX711), servo actuation of the valve, and reliable interconnection between Jetson outputs, the FSM, and Arduino control. End-to-end testing will be repeated and documented with photos and measured results.

Deliverable 6 – Final Design Report

Once the prototype is stable, the project will shift toward documentation and validation. The final report will consolidate the complete design (mechanical, electrical, software, FSM), test evidence, calibration results, and reflection on performance versus plan, including updates to the WBS/Gantt and risk mitigation outcomes.

Deliverable 7 – Final Design Presentation

A structured presentation will be prepared based on the final report, focusing on the problem, the final design, the state-machine behaviour, demonstration results, and key engineering decisions. A short demo flow will be rehearsed to ensure consistent and reliable presentation delivery.

Deliverable 8 – Innovation Fair Participation

The prototype will be made demonstration-ready with an emphasis on repeatability and safety. A clear poster narrative and short pitch will be prepared, supported by visuals (block diagram, FSM, test results) and a live demonstration procedure to show detection-to-action behaviour and captured-mass reporting.

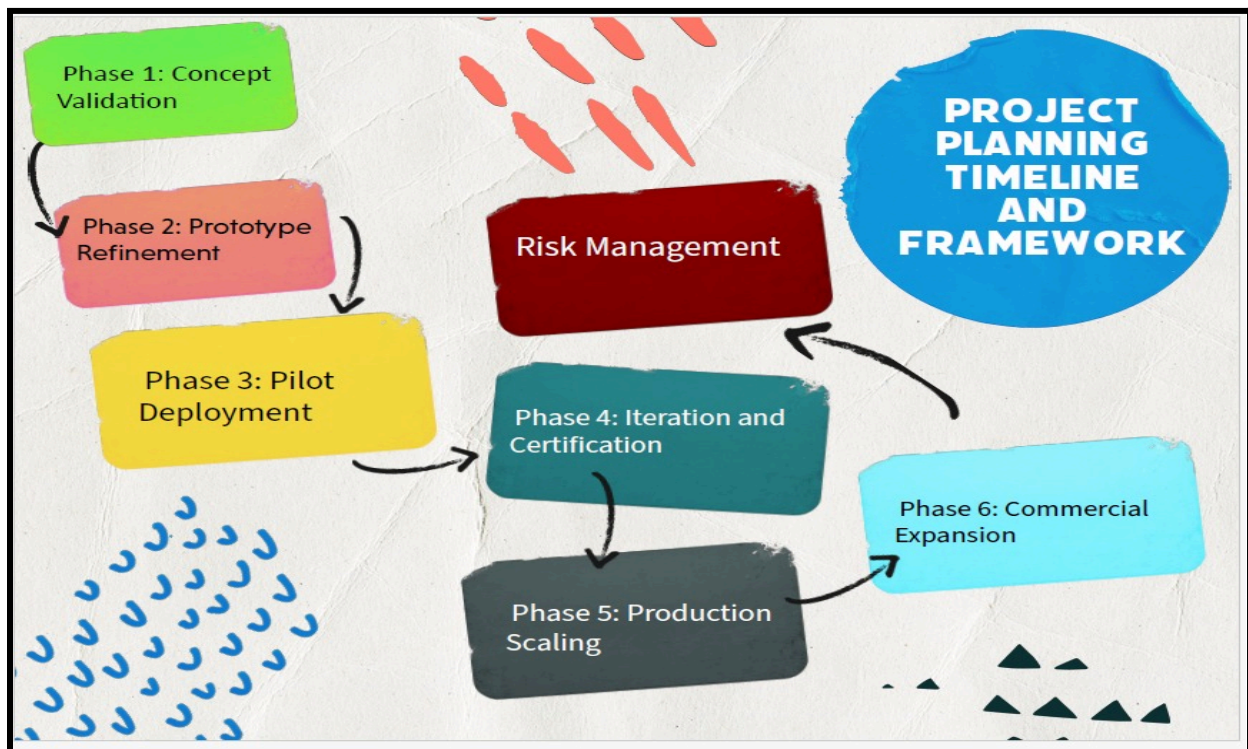


Figure 14 : Flow Chart Showing the Project Planning Time Line and Frame Work

6.2 Gantt Chart

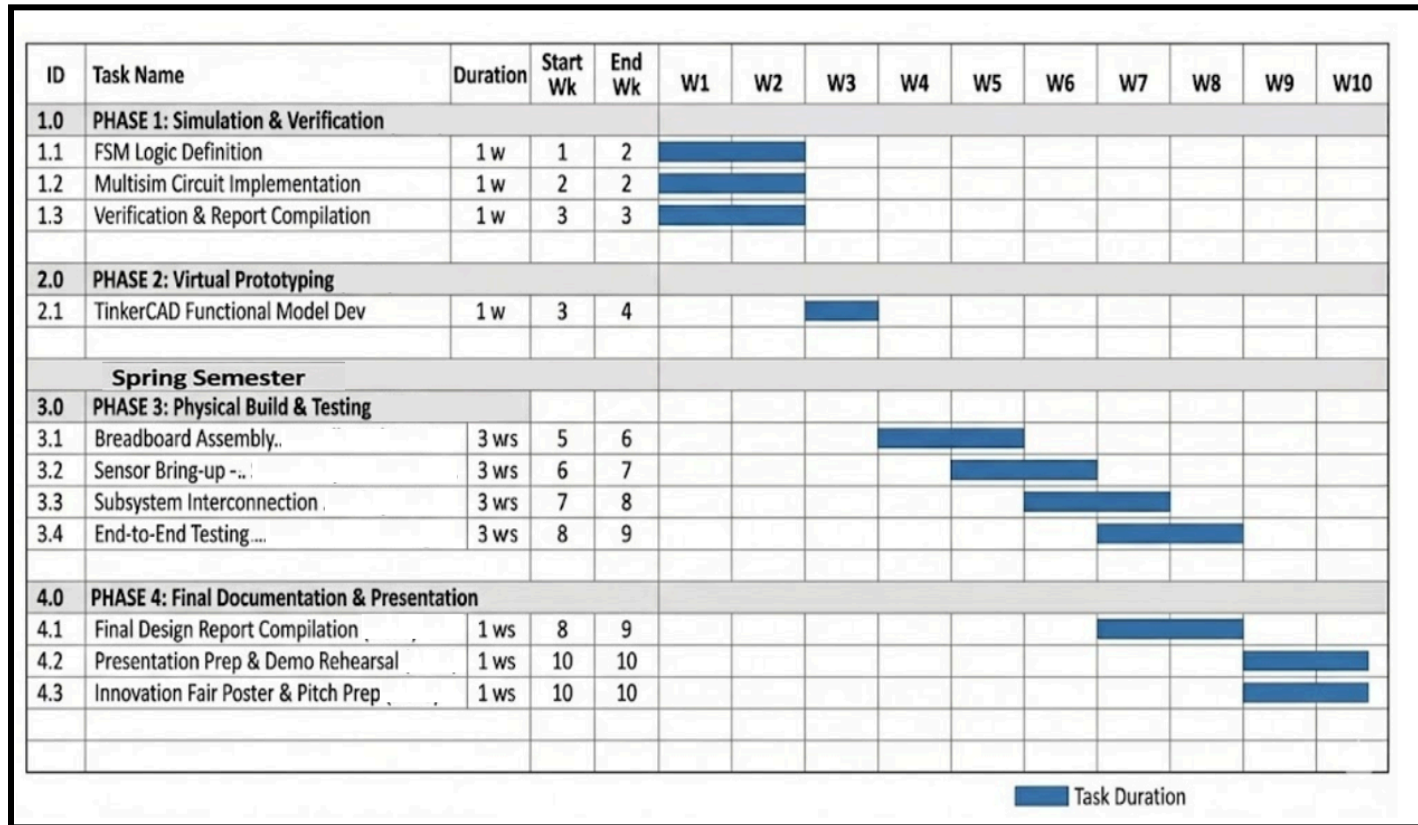


Figure 15 : Gantt Chart

6.3 Work Breakdown Structure (WBS)

The Work Breakdown Structure (WBS) in Figure 16 summarises the project plan by breaking the work into clear work packages and allocating ownership across each team member

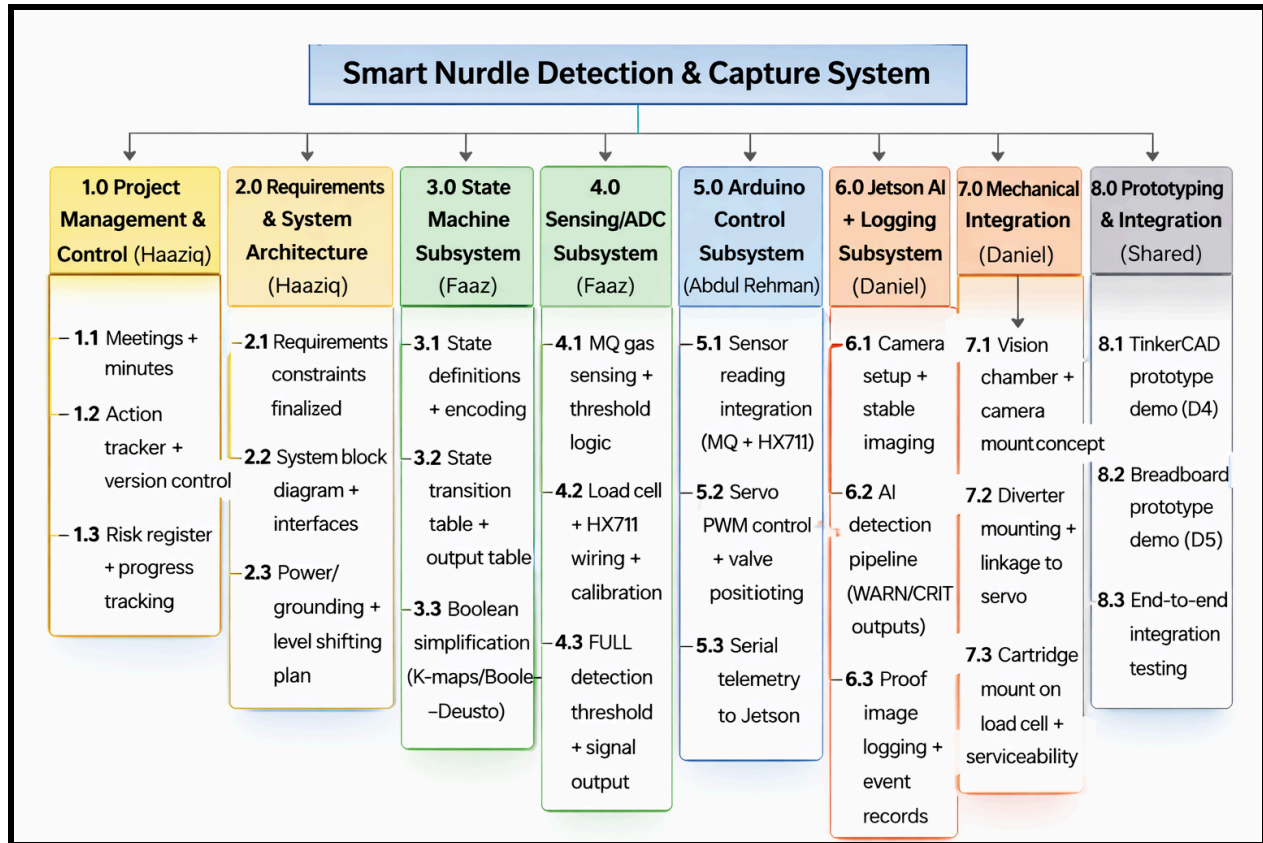


Figure 16 : WBS Chart

6.4 Risk Mitigation

- Integration failures (Jetson–FSM–Arduino not working together): Managed by freezing interfaces early (signal definitions, wiring plan, serial format), using version control, and running incremental end-to-end tests after each subsystem is completed.
- AI detection reliability (false positives/negatives due to bubbles, glare, turbidity): Managed by controlled lighting (UV ring), confidence thresholds with debounce logic, and repeated testing under realistic flow conditions.
- Sensor accuracy and drift (load cell and gas sensor giving unstable readings): Managed by calibration routines, filtering and hysteresis thresholds, and repeat measurements with known references to confirm

consistency.

- Mechanical issues (valve linkage, leaks, or cartridge fit affecting capture): Managed by early mechanical mock-ups, repeated actuation tests, and basic serviceability checks before full demonstrations.

7. Marketing

7.1 Business Model

Customer Segments: The primary customer base includes polymer producers, ports, logistics hubs, pellet storage yards, plastic processors, and industrial parks where pellet handling and runoff risks are highest. Secondary customers include regulators, municipalities, and environmental agencies that fund strategic monitoring and compliance initiatives.

Offering Structure: The solution is offered as either a hardware sale or a lease model, supported by software services. The software layer provides dashboard access, event logging, automated reporting, and performance analytics, while service contracts cover installation support, maintenance, calibration, and periodic system health checks.

Partnerships: The implementation strategy relies on key partnerships to support scale and reliability. Industrial automation and sensor partners can supply certified components, telecom partners can provide IoT connectivity where required, cloud partners can support secure and scalable data infrastructure, and operations and maintenance partners can deliver on-site servicing and rapid response support.

Go-to-Market Strategy: Market entry will begin with pilot deployments in high-visibility UAE sites to validate performance and generate credible case studies. These pilots will be measured using clear KPIs such as captured mass, response time, uptime, and incident reduction, and the results will be used to support broader rollout across the GCC followed by expansion into other regions with tightening pellet loss controls

7.2. Financial Projection and Broader Impact

Revenue Model: Revenue is generated through hardware deployment and recurring subscriptions. Predictable recurring revenue is achieved through multi-site monitoring services.

Customer ROI: The device pays back by preventing even one moderate incident that would otherwise require cleanup, downtime, and reputational repair. The greatest value is avoiding long, incomplete remediation after dispersal.

Scaling Economics: As deployment scales, per-unit manufacturing cost decreases, and software margins improve, enabling competitive pricing and faster adoption.



Figure 17 : Graphical Flow Chart of the Business Model

8. Budget

Table 2: Budget Distribution

Item	Subtotal (AED)
Jetson Nano	550
MicroSD + heatsink/fan	100
Camera	120
UV LED ring + driver	60
Arduino Uno/Nano	60
Load cell (1–5 kg)	45
HX711 amplifier module	20
MQ gas sensor module (MQ-135/MQ-4)	25
High torque servo (MG996R)	45
Y-diverter / valve body	120
Acrylic/PLA/PETG enclosure materials	120
Wires, connectors, resistors, LEDs, buzzer	80
5V 4A supply + buck/aux regulators	85
Total Cost	1430 AED

Without the Jetson Nano, the total cost is 880 AED (since the Jetson Nano is provided by the university)

9. Social Impact

9.1 Connection to the UAE Environment

The UAE's petrochemical industry, high-volume pellet logistics, and major ports/free zones increase the likelihood of pellet loss during loading, handling, and yard operations, while the coastline remains economically and environmentally critical. Because the Arabian Gulf is semi-enclosed with limited water exchange, microplastics can persist and accumulate in sediments and nearshore habitats, making prevention at drains and outfalls essential. Although "zero pellet loss" programs and ESG targets are growing, voluntary housekeeping alone can fail during heavy rainfall, peak operations, or contractor-heavy schedules. Nurdle DNA strengthens UAE readiness by converting zero-loss commitments into measurable, auditable outcomes through real-time monitoring and automatic containment. It supports green port/industrial hub goals and protects beaches and tourism by stopping small leaks before they become public incidents. It also aligns with the UAE's smart infrastructure direction by integrating IoT-based environmental and safety monitoring into under-instrumented drainage systems.

The project Aligns with UAE Policies and Strategy. The UAE Water Security Strategy 2036 emphasises protecting water resources and strengthening water management so water remains safe and reliable, including supporting better monitoring and control of water quality. This project aligns by providing real-time detection and interception of microplastics (nurdles) in drains/outfalls, with automated diversion/capture and audit logs that reduce pollution before it reaches the marine environment or reuse networks. At the same time, the UAE Circular Economy Policy 2021–2031 supports reducing resource loss and environmental leakage; the device contributes by preventing pellet losses at source and generating measurable data (captured mass, event frequency) that drives continuous improvement in industrial handling and compliance.



Figure 18(a) :UAE Water Security Strategy 2036 (b) : UAE Circular Economy Policy

9.2 SDG Goals

Nurdle DNA aligns strongly with several United Nations Sustainable Development Goals, particularly SDG 6 (Clean Water and Sanitation), SDG 12 (Responsible Consumption and Production), and SDG 14 (Life Below Water). By preventing microplastic pellets from entering drainage systems and coastal waters, the system protects marine ecosystems and supports water quality improvement. Its real-time monitoring, automatic containment, and evidence-based reporting promote responsible industrial practices and waste reduction. Additionally, the IoT-based design supports SDG 9 (Industry, Innovation and Infrastructure) by modernizing environmental monitoring infrastructure and enhancing sustainable industrial development and SDG 3 Good Health and Well Being.



Figure 19: Sustainable Development Goals

10. Appendix

10.1 Arduino Code

```
#include <Servo.h>

#include <Adafruit_LiquidCrystal.h>

// --- PIN DEFINITIONS ---

const int PIN_LDR = A0;    // Microplastic Sensor

const int PIN_GAS = A1;    // Gas Sensor

const int PIN_CHEM = A2;   // Chemical (Potentiometer)
```

```

const int PIN_RESET = 2;    // Reset Button

const int PIN_GREEN = 5;    // Normal LED

const int PIN_YELLOW = 6;   // Warning LED

const int PIN_RED = 7;      // Alarm LED

const int PIN_BUZZER = 8;   // Alarm Buzzer

const int PIN_SERVO = 9;    // Valve Servo


// --- THRESHOLDS (Adjust these based on simulation tests) ---

const int THRESH_LDR_ALARM = 800; // Darkness level (turbidity)

const int THRESH_GAS_ALARM = 600; // Gas concentration

const int THRESH_CHEM_ALARM = 900; // pH/Conductivity deviation


// --- STATE VARIABLES ---

bool isLatchedAlarm = false;

Servo valveServo;

Adafruit_LiquidCrystal lcd(0); // Set up LCD at default address 0


void setup() {

    // Setup Pins

    pinMode(PIN_GREEN, OUTPUT);

    pinMode(PIN_YELLOW, OUTPUT);

    pinMode(PIN_RED, OUTPUT);

    pinMode(PIN_BUZZER, OUTPUT);

```

```

pinMode(PIN_RESET, INPUT);

// Setup Servo

valveServo.attach(PIN_SERVO);

valveServo.write(0); // Start OPEN


// Setup LCD

lcd.begin(16, 2);

lcd.setBacklight(1);

lcd.print("Nurdle DNA");

lcd.setCursor(0, 1);

lcd.print("System Init...");

delay(2000); // Show intro screen

lcd.clear();

}

void loop() {

    // 1. READ SENSORS

    int valLDR = analogRead(PIN_LDR);

    int valGas = analogRead(PIN_GAS);

    int valChem = analogRead(PIN_CHEM);


    // 2. CHECK FOR HAZARDS

    bool alarmCondition = (valLDR > THRESH_LDR_ALARM) ||

```

```

        (valGas > THRESH_GAS_ALARM) ||

        (valChem > THRESH_CHEM_ALARM);

// 3. STATE MACHINE LOGIC

if (isLatchedAlarm) {

    // --- ALARM STATE (Latched) ---

    digitalWrite(PIN_RED, HIGH);

    digitalWrite(PIN_GREEN, LOW);

    digitalWrite(PIN_YELLOW, LOW);

    tone(PIN_BUZZER, 1000);

    valveServo.write(90); // CLOSE VALVE


// LCD Feedback

    lcd.setCursor(0, 0);

    lcd.print("HAZARD DETECTED!");

    lcd.setCursor(0, 1);

    lcd.print("Valve CLOSED  ");


// Check Reset

if (digitalRead(PIN_RESET) == HIGH) {

    isLatchedAlarm = false;

    noTone(PIN_BUZZER);

    lcd.clear();

```

```

    lcd.print("System Resetting");

    delay(1000);

    lcd.clear();

}

}

else if (alarmCondition) {

    // --- TRIGGER ALARM ---

    isLatchedAlarm = true;

}

else {

    // --- NORMAL STATE ---

    digitalWrite(PIN_GREEN, HIGH);

    digitalWrite(PIN_RED, LOW);

    digitalWrite(PIN_YELLOW, LOW);

    noTone(PIN_BUZZER);

    valveServo.write(0); // OPEN VALVE


    // LCD Feedback

    lcd.setCursor(0, 0);

    lcd.print("System Normal ");

    lcd.setCursor(0, 1);

    // Display one sensor value (e.g., Turbidity)

    lcd.print("Turbidity: ");

```

```

    lcd.print(valLDR);

}

delay(200); // Update speed

}

```

10.2 Team Role Contribution

Table 3: Team Contribution

Team Member Name	Report Development	System Design
Muhammad Haaziq	Brainstormed the Detail Report Design and Structure Designed Flow Charts , Block Chart and Gantt Chart , Social Impact (SDG and UAE Strategy) and Added Images, Graphs and Mapping Statics for the Report	System Design Architect and Looked after Device Constrain Designed the 3D Model Sketch of the Prototype and its Implementation in Industrial Environment
Daniel Koshy	Developed the Detail Report Design and Structure Added Table for Finance and WBS Chart	Circuit Simulation (Multisim) Designed the Simulation and Device Diversification and its Implementation
Faaz Ali Sayyed	Added Technical aspect of the circuit	Circuit Simulation (Thinker Cad) Designed State Diagram
Mohammed Abdul Rahman	Marketing and Managed Meeting Time for Report	Looked after constraints of the Project

10.3 Team Meeting Minutes

Meeting 1

Person In Charge : Muhammad Haaziq (Managing Director)

Attendees in Meeting : Daniel , Faaz and Mohammed Abdul Rahman

Date : 8 th January 2026

Time: 8:30 -10:30 PM

Session Type: Online Team Meeting

Duration: 2 hour

Goal for this session:To Determine the Idea for the Project for Deliverable 1

Activities done: Discussed many Ideas in Physics , Medicine and Environmental Field of Engineering for the Project

Decisions made:**Planned** on idea called The "Stabil-X" Haptic Telepresence System and the Nurdle DNA Themes: Industry 4.0, Humanitarian Engineering, Health (Assistive Tech)

Meeting 2

Person In Charge : Muhammad Haaziq (Managing Director)

Attendees in Meeting : Daniel , Faaz and Mohammed Abdul Rahman

Date : 15 th January 2026

Time: 12:30 -2:30 PM

Session Type: Team Meeting in the University

Duration: 2 hour

Goal for this session:To Finalize the Idea for the Project for Deliverable 1 which is Nurdle DNA

Activities done: Discussed many Ideas in Physics , Medicine and Environmental Field of Engineering for the Project

Decisions made: Finalized the Nurdle DNA which comes under the Themes: Industry 4.0, Humanitarian Engineering, and Health

11. References

The Great Nurdle Hunt, n.d. The Great Nurdle Hunt. Fidra. Available at: <https://www.nurdlehunt.org.uk/>.

United Nations (n.d.) The 17 Goals. Available at: <https://sdgs.un.org/goals>

Ali, T., Mortula, M., Mohsen, B., Dronjak, L., Gawai, R., Atabay, S., Khan, Z. and Fattah, K. (2024) ‘Evaluating Microplastic Pollution Along the Dubai Coast: An Empirical Model Combining On-Site Sampling and Sentinel-2 Remote Sensing Data’, *Journal of Sustainable Development of Energy, Water and Environment Systems*, 12(1), 1110482. doi:10.13044/j.sdewes.d11.0482

Pu, S., Bushnaq, H., Munro, C., Gibert, Y., Sharma, R., Mishra, V. and Dumée, L.F. (2024) *Perspectives on transport pathways of microplastics across the Middle East and North Africa (MENA) region*, npj Clean Water, 7(1), Article 114. doi:10.1038/s41545-024-00410-w.

National Oceanic and Atmospheric Administration (NOAA) Ocean Service (n.d.) What are microplastics? Available at: <https://oceanservice.noaa.gov/facts/microplastics.html>

World Health Organization (WHO) (2019) Microplastics in drinking-water. Available at: <https://www.who.int/publications/i/item/9789241516198>

Operation Clean Sweep (n.d.) Home. Available at: <https://opcleansweep.org/>

Gulf Petrochemicals and Chemicals Association (GPCA) (n.d.) GPCA Operation Clean Sweep (OCS) Scheme. Available at: <https://gpcachem.org/operation-clean-sweep-2/gpca-operation-clean-sweep-ocs-scheme/>

NVIDIA Developer (n.d.) Get Started With Jetson Nano Developer Kit. Available at: <https://developer.nvidia.com/embedded/learn/get-started-jetson-nano-devkit>

National Institute of Standards and Technology (NIST) (n.d.) ‘finite state machine’, Dictionary of Algorithms and Data Structures. Available at: <https://xlinux.nist.gov/dads/HTML/finiteStateMachine.html>

Avia Semiconductor (n.d.) HX711: 24-Bit Analog-to-Digital Converter (ADC) for Weigh Scales [PDF]. Available at:

https://cdn.sparkfun.com/datasheets/Sensors/ForceFlex/hx711_english.pdf

Zhengzhou Winsen Electronics Technology Co., Ltd. (n.d.) MQ135 Semiconductor Sensor for Air Quality (Ver. 1.4) Manual [PDF]. Available at:

<https://www.winsen-sensor.com/d/files/PDF/Semiconductor%20Gas%20Sensor/MQ135%20%28Ver1.4%29%20-%20Manual.pdf>

Watson-Marlow Fluid Technology Solutions (WMFTS) (n.d.) Peristaltic pumps - how do they work? Available at:

<https://www.wmfts.com/en-ae/support/pump-principles/how-do-peristaltic-pumps-work/>